

# LAEEQ AHMED

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## PROFESSIONAL SUMMARY

Results-driven **Software & AI Engineer** with hands-on experience in building scalable full-stack applications and AI systems. Proficient in **Python**, the **MERN stack**, and **FastAPI**, with strong **DSA** and **OOP** fundamentals. Skilled in designing RESTful APIs, implementing **Retrieval-Augmented Generation (RAG)**, and managing **PostgreSQL/MongoDB**. Proven ability to deploy containerized ML and backend solutions using **Docker**, **AWS**, and CI/CD pipelines.

## EDUCATION

**Matrusri Engineering College (Osmania University)**

Bachelor of Engineering (B.E.) in Information Technology | CGPA: 8.23/10

Hyderabad, India

Nov 2022 – Jun 2026

## TECHNICAL SKILLS

**Programming Languages:** C++, Python, Java, JavaScript, SQL

**Full Stack & CS Fundamentals:** React.js, Node.js, Express.js, FastAPI, HTML5, CSS3, RESTful APIs, JWT Auth, OOP, DSA

**AI/ML & Frameworks:** PyTorch, Tensorflow, LangGraph, LangChain, RAG, LLMs

**Databases:** PostgreSQL, MongoDB, MySQL, Vector Databases (ChromaDB)

**Cloud & DevOps:** AWS, Docker, CI/CD (GitHub Actions), Git, Unit Testing, Agile Methodologies, SDLC

## EXPERIENCE

**Swecha & IIIT Hyderabad (Viswam AI)**

Hyderabad, India

AI Developer Intern

Jun 2025 – Jul 2025

- Collaborated with a cross-functional team in an **Agile** environment to procure, clean, and anonymize 5,000+ SQL queries, establishing a secure dataset for local LLM fine-tuning.
- Engineered an autonomous Text-to-SQL agent workflow using **LangGraph** and **Python**, integrating a fine-tuned **Llama-3 (8B)** model via **QLoRA** to translate natural language into optimized database queries.
- Implemented structured evaluation workflows to achieve **92% execution accuracy**; built a **FastAPI** backend containerized with **Docker** to enable automated testing and seamless deployment.

## PROJECTS

**CareerX AI – Full-Stack AI Career Intelligence Platform**

Live Demo | GitHub

React.js, Node.js, Express.js, Python, MongoDB, ChromaDB, RAG, Docker, CI/CD

- Architected a scalable full-stack application utilizing the **MERN stack** and **RAG** to deliver 4 core generative modules (mock interviews, ATS analysis, 1-click resume customization), processing context-aware outputs in **under 3 seconds**.
- Applied **OOP** principles to design a robust backend architecture, developing **15+ RESTful APIs** using **Node.js** and **Express.js** for secure authentication (OAuth/JWT) and complex payment workflows.
- Built a distributed storage system using **MongoDB** for user data and **ChromaDB** to store high-dimensional text embeddings, achieving vector search latency for accurate semantic similarity scoring.
- Containerized AI microservices and database instances using **Docker**, and automated the deployment lifecycle via **GitHub Actions** to accelerate release cycles by **30%** and maintain a reliable CI/CD pipeline.

**SmartFirePredict – Real-Time Computer Vision & IoT System**

Live Demo | GitHub

Python, PyTorch, FastAPI, YOLOv8, OpenCV, ESP32, Docker, AWS

- Developed an end-to-end AI/IoT fire detection system from scratch using **Python**, **YOLOv8** (achieving an **87% mAP50** score), and **ESP32** microcontrollers.
- Designed a multi-threaded **FastAPI** inference server capable of processing concurrent RTSP and WebSocket video streams, optimizing backend logic to reduce end-to-end latency by **1-2 seconds**.
- Engineered multimodal sensor fusion logic combining computer vision bounding boxes with hardware data to reduce false alarms by **20-30%**; integrated Twilio/Telegram APIs for real-time alerts.
- Packaged deep learning models into **Docker** containers, implemented **unit tests** for system reliability, and deployed the highly available system to **AWS EC2** using robust CI/CD pipelines.

## LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

**Tech Tycoons Club, Matrusri Engineering College**

Hyderabad

Tech Executive

2024 – Present

- Organized campus-wide hackathons and coding contests, managing cross-functional student teams.
- Conducted hands-on workshops on **Programming Fundamentals**, **DSA**, **Python**, and **Git/GitHub** workflows to support peer learning and technical development.

## ACHIEVEMENTS

- Achieved **Global Rank 1130** in **TCS CodeVita Season 13**, demonstrating exceptional **problem solving**, **DSA**, and algorithmic complexity analysis skills under strict time constraints.
- Secured **First Prize** in **Inter-Department Major Project Presentation**, articulating architecture, technical challenges, and software solutions to a panel of judges.
- Awarded **First Runner-Up** in **Sprint Hackathon**, showcasing the ability to handle ambiguous technical challenges and rapidly deliver functional prototypes.